

Zisen Dai

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Education

Tsinghua University

Bachelor of Science in Physics | GPA: 4.0/4.0 | Rank: 1/28

August 2023 – Present

Beijing, China

UC Berkeley

Exchange Student in Astrophysics | GPA: 4.0/4.0 | Adviser: Wenbin Lu

January 2026 – May 2026

Berkeley, CA, USA

Research Experience

Department of Astronomy, UC Berkeley

Research Assistant (Advised by Wenbin Lu)

May 2026 – Present

Berkeley, CA, USA

- Tracking cutting-edge developments in high-energy transients via weekly group discussions.

Institute for Advanced Study, Tsinghua University

Research Assistant (Advised by Xuening Bai)

March 2025 – Present

Beijing, China

- Investigating cosmic ray transport dynamics within protoplanetary disk (PPD) winds.
- Developed a Python simulation framework from scratch to model particle transport via the Guiding Center Approximation, analyzing the impact of scattering mechanisms and strengths.

Project

Rosseland-Mean Opacity Computation

Calculate the Rosseland-Mean Opacity of Pure Hydrogen Gas under LTE (report, code)

April 2026

Berkeley, CA, USA

- Developed a Python framework to computing the Rosseland-mean opacity of pure hydrogen gas in LTE, and benchmarking it against the LANL/TOPS opacity tables.
- Localized the main opacity discrepancy domains and analyzed the missing physical channels and omitted physics causing the issue.

LBGs Selection Based on Photometric Data

Machine Learning-Based High-Redshift Target Selection (report, code)

January 2026

Beijing, China

- Developed a LightGBM classifier on CLAUDS photometry, utilizing sample weighting and probability calibration to identify high-redshift ($z \sim 2 - 3$) Lyman Break Galaxies.
- Corrected selection bias via TSR/SSR metrics, achieving 0.995 AUC and 92.0% purity — outperforming the traditional u -dropout method with a $3.3\times$ sample size increase.

JUNO Optical Simulation & Signal Reconstruction

Comprehensive Physics Simulation and Machine Learning Modeling (code)

August 2025

Beijing, China

- Developed a Monte Carlo Python framework to simulate complex photon transport, Fresnel reflection, and emission for 17,612 PMTs.
- Engineered an ultra-high-capacity LightGBM model with multi-center Gaussian attention to reconstruct spatial-temporal Probe functions, reducing temporal integral error to $\sim 3\%$.

Spectrum and Light Curve Analysis of Supernovae

Photometric and Spectroscopic Classification (report, code)

June 2025

Beijing, China

- Classified SN 2025kid as a Type Ia supernova by analyzing multi-band photometric data and spectroscopic absorption features.
- Extracted key parameters (peak magnitude, decline rate Δm_{15}) of SN 2025fvw via light curve fitting, utilizing the Phillips relation for standardization.

Awards & Honors

National Scholarship <i>Highest Honor for Undergraduates, The Ministry of Education of China</i>	2024, 2025
Tsinghua Xuetaang Talents Program Scholarship <i>Chi-Sun Yeh Class, Elite Physics Track</i>	2023 – 2025
Excellent Academic Performance Scholarship <i>Tsinghua University</i>	2024, 2025
Basic Disciplines Interdisciplinary Challenge <i>Third Prize, Tsinghua University</i>	2025
Chinese Physics Olympiad (CPhO) <i>First Prize in Zhejiang</i>	2022

Selected Coursework

Perfect grades (4.00/4.00) across all 26 major courses (including **8 A+**, **14 A**)

Astrophysics: Relativistic Astrophysics and Cosmology (A+), Statistical Methods in Astrophysics (A+), Observational Astronomy (A+), Radiation Processes in Astronomy, Astrophysics

Advanced Physics: Statistical Mechanics (A+), Quantum Mechanics, Electrodynamics, Analytical Mechanics, Atomic and Molecular Physics

Advanced Mathematics: Mathematical Physics Equations (A+), Complex Analysis, Linear Algebra for Physical Sciences, Advanced Calculus I & II

Data: Big Data in Experimental Physics I & II (A+), Introduction to Probability and Statistics (A+)

Other Experience

Volunteer Peer Tutor <i>Drop-in Tutoring Center, Tsinghua University</i>	March 2025 – Present <i>Beijing, China</i>
• Providing one-on-one academic support and problem-solving strategies for undergraduates in advanced physics and math courses.	
Student Life Coordinator <i>Department of Organization, Zhili College</i>	March 2024 – December 2025 <i>Beijing, China</i>
• Organized college-wide student events, coordinated group activities, and facilitated team-building initiatives.	
Academic Representative <i>Physics Class 32, Zhili College</i>	September 2023 – September 2025 <i>Beijing, China</i>
• Liaised between faculty and students to facilitate academic communications and organize study groups.	